PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

28 Dec – 3 Jan 2027 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
28 Dec D2026-12-28	Daily mixed Daily test	___ / 10	Retest: 31 Dec
29 Dec D2026-12-29	Daily mixed Daily test	___ / 10	Retest: 1 Jan
30 Dec D2026-12-30	Daily mixed Daily test	___ / 10	Retest: 2 Jan
3 Jan T2027-01-03	Science Weekly test	___ / 40	Retest: 6 Jan

Daily Test · Answer key

Monday, 28 December 2026 · 10 marks

Test ID: D2026-12-28

Science Ch 2 Cell · Maths Ch 1 Coordinates · AI Green Skills · Maths Ch 14 Surface Area & Volume

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Q1 [1]

Question: Which organelle is called the “powerhouse of the cell”?

Expected answer / marking points: Mitochondria.

Q2 [2]

Question: Write two differences between a plant cell and an animal cell.

Expected answer / marking points: Plant cell has a cell wall, animal cell does not [1]; plant cell has chloroplasts and a large vacuole, animal cell has small/no vacuoles and no chloroplasts [1].

Q3 [1]

Question: Write the coordinates of a point on the x-axis at a distance of 6 units to the left of the origin.

Expected answer / marking points: (–6, 0).

Q4 [2]

Question: Find the distance between (–2, –5) and (–2, 4). In which quadrant does (–2, –5) lie?

Expected answer / marking points: $|4 - (-5)| = 9$ units [1]; III quadrant [1].

Q5 [1]

Question: What is a green job?

Expected answer / marking points: A job that helps protect or restore the environment, e.g., solar technician, recycling worker.

Q6 [1]

Question: Find the surface area of a sphere of radius 7 cm.

Expected answer / marking points: $4 \times 22/7 \times 49 = 616$ cm^2.

Q7 [2]

Question: Find the slant height of a cone with radius 6 cm and height 8 cm, and its curved surface area ($\pi = 3.14$).

Expected answer / marking points: $l = \sqrt{(36 + 64)} = 10$ cm [1]; CSA $= 3.14 \times 6 \times 10 = 188.4$ cm^2 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Tuesday, 29 December 2026 · 10 marks

Test ID: D2026-12-29

SST Ch 2 Earth's surface · English Grammar + Writing · Science Ch 11 Reproduction · Maths Ch 14 Surface Area & Volume

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Q1 [1]

Question: Name the three main layers of the Earth.

Expected answer / marking points: Crust, mantle, core.

Q2 [2]

Question: Distinguish between internal (endogenic) and external (exogenic) forces with one example of each.

Expected answer / marking points: Internal forces act from inside and build/move the crust — earthquakes, volcanoes, folding [1]; external forces wear the surface down — weathering, erosion by rivers, wind, glaciers [1].

Q3 [2]

Question: Change into reported speech: Grandpa said to Ravi, "Set out the chessboard."

Expected answer / marking points: Grandpa told/asked Ravi to set out the chessboard [2].

Q4 [2]

Question: Correct the errors: (i) He don't like tea. (ii) She has went to school.

Expected answer / marking points: (i) He doesn't like tea [1]. (ii) She has gone to school [1].

Q5 [1]

Question: What is formed when a male gamete fuses with a female gamete?

Expected answer / marking points: Zygote.

Q6 [2]

Question: A cuboidal box is 50 cm × 40 cm × 30 cm. How many cubes of edge 10 cm can it hold?

Expected answer / marking points: Volume ratio: 60000 ÷ 1000 [1] = 60 cubes [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 30 December 2026 · 10 marks

Test ID: D2026-12-30

Hindi पाठ 4 ऐसी भी बातें होती हैं · Maths 5.1–5.2 Circle basics · SST P2 Ch 3 Resistance & Resilience · Maths Ch 14 Surface Area & Volume

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Q1 [2]

Question: लता मंगेशकर के जीवन से हमें कौन-सी दो प्रेरणाएँ मिलती हैं?

Expected answer / marking points: संघर्षों के बीच भी कला के प्रति समर्पण [1]; परिवार के प्रति ज़िम्मेदारी निभाना (या निरंतर अभ्यास) [1]।

Q2 [1]

Question: A circle has rotational symmetry about its _____ through any angle.

Expected answer / marking points: centre

Q3 [1]

Question: How many lines of reflection symmetry does a circle have?

Expected answer / marking points: Infinitely many — every diameter is a line of symmetry.

Q4 [1]

Question: Who led the Mewar forces at the Battle of Haldighati (1576)?

Expected answer / marking points: Maharana Pratap.

Q5 [1]

Question: Who defeated the Mughal fleet at the Battle of Saraighat (1671)?

Expected answer / marking points: The Ahom commander Lachit Borphukan.

Q6 [2]

Question: Find the volume of a cylinder of radius 7 cm and height 10 cm.

Expected answer / marking points: 22/7 × 49 × 10 [1] = 1540 cm^3 [1].

Q7 [2]

Question: Find the volume of a sphere of radius 3 cm (π = 3.14).

Expected answer / marking points: 4/3 × 3.14 × 27 [1] = 113.04 cm^3 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Science · Weekly Test 2 · Answer key

Sunday, 3 January 2027 · 40 marks

Test ID: T2027-01-03

Class 9 · Ch 7 Work & Energy · Ch 9 Atomic Foundations · Ch 10 Sound · Ch 11 Reproduction · Ch 12 Diversity + revision Ch 4, 6

Q1 [1]

Question: The SI unit of power is:

(a) joule (b) newton

(c) watt (d) pascal

Expected answer / marking points: (c) watt

Q2 [1]

Question: A man pushes a wall but it does not move. The work done by him on the wall is:

(a) positive (b) negative

(c) zero (d) maximum

Expected answer / marking points: (c) zero

Q3 [1]

Question: The molecular mass of H_2SO_4 (H = 1, S = 32, O = 16) is:

(a) 49 u (b) 96 u

(c) 98 u (d) 100 u

Expected answer / marking points: (c) 98 u

Q4 [1]

Question: Which of the following is formed by transfer of electrons?

(a) Cl_2 (b) H_2O

(c) NaCl (d) CH_4

Expected answer / marking points: (c) NaCl

Q5 [1]

Question: If 20 compressions pass a point in 4 seconds, the frequency of the sound is:

(a) 80 Hz (b) 5 Hz

(c) 10 Hz (d) 0.2 Hz

Expected answer / marking points: (b) 5 Hz

Q6 [1]

Question: Sound cannot travel through:

(a) steel (b) water

(c) air (d) vacuum

Expected answer / marking points: (d) vacuum

Q7 [1]

Question: Bryophyllum reproduces vegetatively by:

(a) roots (b) leaf buds

(c) spores (d) seeds

Expected answer / marking points: (b) leaf buds

Q8 **Question:** Which kingdom has multicellular organisms that are heterotrophic and have a cell wall made of chitin? **[1]**

(a) Monera

(b) Protista

(c) Fungi

(d) Plantae

Expected answer / marking points: (c) Fungi

Q9 **Question:** A car starts from rest and reaches 24 m s^{-1} in 6 s. Its acceleration is: **[1]**

(a) 4 m s^{-2} (b) 6 m s^{-2}

(c) 144 m s^{-2} (d) 0.25 m s^{-2}

Expected answer / marking points: (a) 4 m s^{-2}

Q10 **Question: A:** When a ball thrown upwards reaches its highest point, its potential energy is maximum. **[1]** **R:** At the highest point, the velocity of the ball is zero, so all its kinetic energy has changed into potential energy.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q11 **Question: A:** Offspring produced by asexual reproduction are genetically identical to the parent. **[1]** **R:** Asexual reproduction involves only one parent and no fusion of gametes.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a)

Q12 **Question:** A boy of mass 40 kg runs up 5 m high stairs in 8 s. Find the power developed by him. **[2]**

Expected answer / marking points: W = mgh = 40 × 10 × 5 = 2000 J [1]; P = 2000/8 = 250 W [1].

Q13 **Question:** Write the chemical formulae of (i) aluminium oxide (ii) calcium hydroxide. **[2]**

Expected answer / marking points: (i) Al_2O_3 [1] (ii) $Ca(OH)_2$ [1].

Q14 **Question:** In a room, reflected sound reaches the ear 0.05 s after it is produced. Will it be heard as an echo or cause reverberation? Justify. **[2]**

Expected answer / marking points: Echo needs at least 0.1 s gap [1]; 0.05 s is less, so it mixes with the original sound — reverberation [1].

Q15 [2] **Question:** Differentiate between self-pollination and cross-pollination. Which gives more variation?

Expected answer / marking points: Self: pollen to stigma of same flower/plant; cross: to a flower of another plant of the same species [1]; cross-pollination gives more variation [1].

Q16 [3] **Question:** A ball of mass 0.2 kg is thrown upward with a speed of 10 m s^{-1}. Find (i) its initial kinetic energy (ii) the maximum height it reaches (iii) its potential energy at that height.

Expected answer / marking points: (i) ½ × 0.2 × 100 = 10 J [1] (ii) mgh = 10 ⇒ h = 10/(0.2 × 10) = 5 m [1] (iii) PE = 10 J (all KE becomes PE) [1].

Q17 [3] **Question:** Element A has 1 electron in its third shell and element B has 6 electrons in its second shell. (i) What ions do A and B form? (ii) What type of bond forms between them? (iii) Write the formula of the compound.

Expected answer / marking points: A (2, 8, 1) = Na loses 1 e^- → A^+; B (2, 6) = O gains 2 e^- → B^{2-} [1]; ionic bond [1]; A_2B (Na_2O) [1].

Q18 [3] **Question:** A sound wave has a frequency of 2 kHz and wavelength 17 cm. (i) Find its speed. (ii) How long will it take to travel 1.7 km? (iii) Name the characteristic that decides pitch.

Expected answer / marking points: (i) v = fλ = 2000 × 0.17 = 340 m s^{-1} [1] (ii) t = 1700/340 = 5 s [1] (iii) Frequency [1].

Q19 [3] **Question:** Why are viruses not placed in any of the five kingdoms? Write two rules of binomial nomenclature.

Expected answer / marking points: Viruses are not cellular and show life only inside a host cell [1]; genus capital, species small letter [1]; written in italics / underlined separately [1].

Q20 [4] **Question:**

> A farmer wants to grow a large number of banana and sugarcane plants that are exactly like the best plants on his farm, and he wants them quickly. His friend suggests growing them from seeds instead.

(i) Which method of reproduction should he use for sugarcane? [1]
(ii) Name one more plant grown commonly this way. [1]
(iii) Give two reasons why this method is better for the farmer than growing from seeds, and one disadvantage of it. [2]

Expected answer / marking points: (i) Vegetative propagation (stem cuttings) [1]. (ii) Any one: rose, potato, grapes, money plant [1]. (iii) Identical to the parent; faster flowering/fruiting; works for seedless plants [1]; disadvantage: no variation, so a disease can affect all plants [1].

Q21 [5] **Question:** State the law of conservation of mechanical energy. A 2 kg stone is dropped from a height of 20 m. Find its potential energy, kinetic energy and speed at (i) the top (ii) 5 m above the ground (iii) just before touching the ground.

ATHVA / OR

(i) State the law of constant proportions. (ii) Explain how a covalent bond is formed with the example of a methane molecule. (iii) Write two differences between ionic and covalent compounds.

Expected answer / marking points: Statement: in the absence of friction, total mechanical energy (PE + KE) stays constant [1]. (i) Top: PE = 2 × 10 × 20 = 400 J, KE = 0, v = 0 [1]. (ii) At 5 m: PE = 100 J, KE = 300 J [1]; v = $\sqrt{(2 \times 300/2)} = \sqrt{300} \approx 17.3$ m s^{-1} [0.5]. (iii) PE = 0, KE = 400 J [1]; v = 20 m s^{-1} [0.5].
OR: (i) A pure compound always contains the same elements in the same fixed proportion by mass [1]. (ii) C has 4 valence electrons; it shares one with each of 4 H atoms, forming 4 single covalent bonds; C gets 8 and each H gets 2 electrons (dot diagram) [2]. (iii) Ionic: high melting point, conduct when molten/dissolved, formed by transfer; covalent: low melting point, usually poor conductors, formed by sharing (any two) [2].

Score: ___ / 40 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: